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Wireless Telemetry Optimizes Test Duration in Rigless Well Testing

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Abstract

Rigless well testing with monobore completion has become a popular method for multi-zone reservoir testing on land. This paper introduces a new wireless telemetry system which integrates the acoustic signal technology with the wireline and slickline technologies in order to optimize the build-up duration and maximize the reservoir evaluation and improve the well test process.

Rigless testing can utilize slickline, wireline and coiled tubing interventions to perforate, kick-off flow and set gauges with downhole shut-in tool (DHSIT). Traditionally, the downhole gauge data will be acquired in memory mode and retrieved for analysis after the build-up is completed. However, the data interpretation may not allow possible extension of the build-up period. The new solution offers the utilization of acoustic telemetry to read-out the data live from an assembly consisting of DHSIT, gauges and acoustic repeater (deployed via slickline prior to the flowback). This downhole set-up will be able to communicate wirelessly with another repeater deployed via wireline during the build-up period. Then, the data is transmitted through wireline to surface for real-time analysis and interpretation.

The solution building and validation process included the system integrity test in the warehouse and later deployed in two runs. The first run allowed a design upgrade to improve the signal between repeaters above and below the DHSIT. Based on this, it is recommended adding the centralizers to have better metal-to-metal contact to increase the acoustic signal-to-noise ratio.

The second run was successful where the wireless repeater was deployed via wireline after 10 hours from the shut-in time of the DHSIT's valve. It was positioned at 100 ft above the downhole valve and it started downloading all historical data from the gauge memory and kept recording in real-time during 36 hours of build-up. During this time there was no interruption in the data transmission and the data was delivered to petroleum engineers in town for real-time interpretation and decision making. During the pull-out-of-hole, the distance at which the signal would be lost was evaluated. The tool was stopped every 200 ft to check the signal quality and found that the longest distance at which the signal was received at 1,200 ft distance from the valve.

The advantage of integration of the wireless telemetry with conventional slickline and wireline operation optimizes the test duration and provides complete reservoir information. It is also recommended to add a valve control option through wireless telemetry to be able to interrupt an on-going build-up episode

without additional slickline run. Other recommendations are to improve the retrieval process by knowing the pressure differential and also to reduced number of fishing operations.

For the first time, the wireless telemetry was introduced for rigless well testing. It was achieved by adapting the conventional acoustic telemetry used in the DST rig operations and integrating it with slickline and wireline technologies. This is also a step forward in developing real-time controllable DHSIT.

Introduction

Exploration well testing is essential for validating hydrocarbon reserves and understanding reservoir characteristics. This process typically includes perforation, stimulation, initial flowback for well clean-up, followed by a multi-rate flow test and pressure build-up for pressure transient analysis. A temporary completion is installed using Drill Stem Test (DST) tools deployed on tubing, allowing for efficient control and monitoring of downhole activities (Hollaender et al. 2019). While DST remains the safest and most effective method for conducting downhole tests, it traditionally requires a rig to be in place. However, in land operations where exploration activity is high, the need to optimize rig utilization has led to the development of rigless well testing. This approach eliminates the dependency on a rig during the testing phase, enabling operators to accelerate drilling schedules and increase the number of wells drilled.

Rigless Well Testing

Rigless well testing typically involves deploying a monobore completion immediately after drilling and logging the well (Fig. 1). This completion must be engineered to withstand the anticipated downhole and surface conditions, including pressure, temperature and exposure to corrosive fluids. Beyond its mechanical integrity, the completion design must also provide access for various intervention techniques such as coiled tubing, wireline, and slickline operations to enable comprehensive downhole testing and data acquisition. These interventions are critical for executing tasks like zonal isolation, setting and retrieving tools, conducting pressure and temperature surveys, and performing stimulation or clean-up operations, all without the need for a rig.

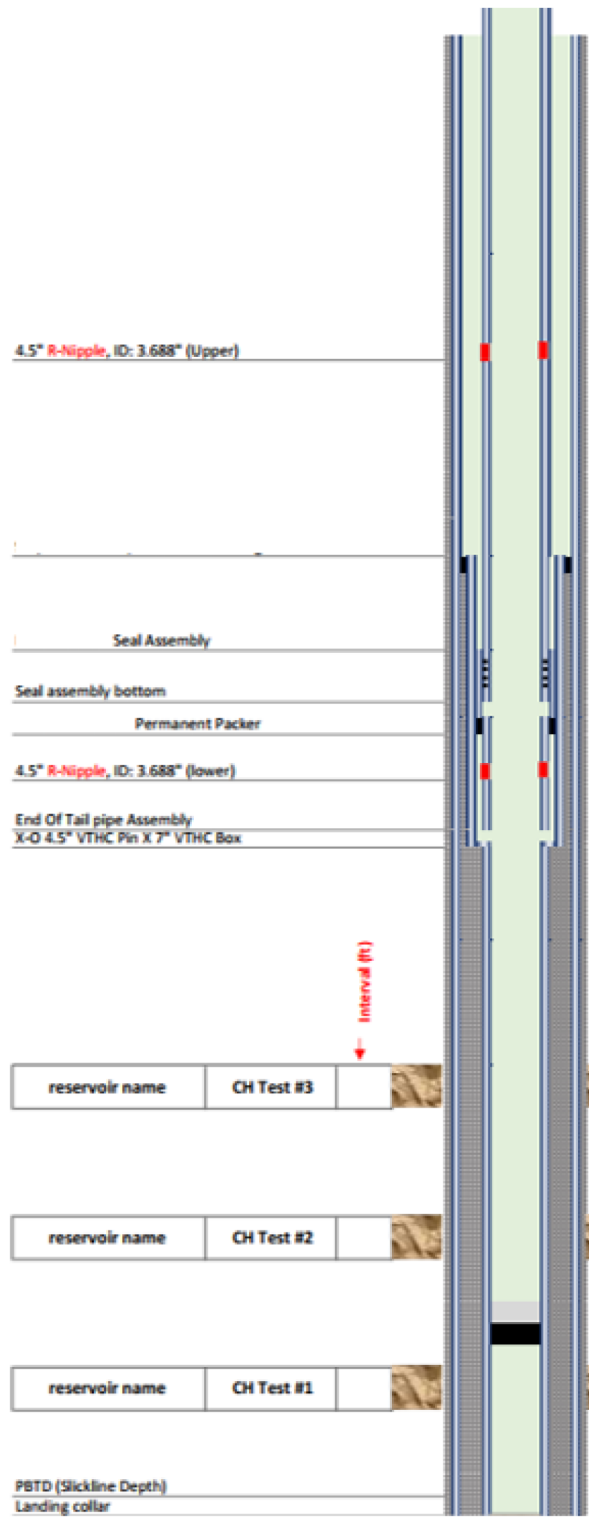


Figure 1—Monobore completion for rigless multi-zone reservoir testing.

Rigless well testing is executed by mobilizing the site (camp), including well testing packages, coiled tubing units, wireline, and slickline services. Each zone is tested individually, starting from the deepest (lowermost) interval and progressing upward (Fig. 1). The typical sequence of downhole operations includes:

1. **Wellbore Displacement** – Brine is pumped into the wellbore using coiled tubing to prepare the well for perforation and testing.
2. **Perforation** – Performed via wireline-conveyed or coiled tubing-deployed perforating guns to access the target reservoir zone.
3. **Gauge Hanger Installation** – A memory gauge hanger is set below the perforated interval using slickline to record downhole pressure and temperature data.
4. **Stimulation** – An acid pumped through coiled tubing to reduce the near-wellbore skin factor and increase the productivity index.
5. **Well Kick-Off** – Nitrogen (N₂) is injected through coiled tubing to initiate flow from the reservoir.
6. **Clean-Up Phase** – The well is allowed to flow either naturally or assisted by N₂ lifting to remove completion fluids and establish stable flow.
7. **Surface Shut-In for DHSIT Installation** – The well is shut in at the surface, and a Downhole Shut-In Tool (DHSIT) equipped with a memory gauge is deployed via slickline.
8. **Main Flow Period** – The well is flowed again, either naturally or with N₂ assistance, to gather production data.
9. **Build-Up Test** – The DHSIT is closed to initiate a pressure build-up period, allowing for reservoir characterization based on pressure transient test method.
10. **Tool Retrieval** – The DHSIT and memory gauge are retrieved from the wellbore using slickline.
11. **Well Kill Operation** – If the acquired downhole data is satisfactory, the well is killed using brine or other kill fluids.
12. **Gauge Hanger Retrieval** – The gauge hanger is pulled out of the hole via slickline.
13. **Zone Isolation and Transition** – The tested zone is isolated using the plug and/or cement bailer run on wireline, and operations proceed to the next zone.

One of the primary limitations of rigless well testing lies in downhole data acquisition during the main build-up, which is typically performed using memory-mode tools. While these tools are effective in capturing pressure and temperature data, they lack real-time transmission capabilities. This can pose challenges in accurately characterizing reservoir heterogeneity, especially when the pressure build-up duration is either too short to capture meaningful data or longer than necessary, leading to extended operational time and increased costs.

In scenarios where pressure stabilizes faster than anticipated, operations may continue unnecessarily, consuming valuable resources. Conversely, if the build-up period is insufficient, critical reservoir features—such as compartmentalization, faults or permeability variations may go undetected.

To address this challenge, this paper introduces an innovative solution aimed at enhancing downhole data acquisition during rigless well testing. The proposed approach seeks to optimize data quality and operational efficiency, enabling more accurate reservoir evaluation without the need for a rig.

Wireless Telemetry with DHSIT

The new solution for rigless downhole data is made of combining existing technologies such as acoustic telemetry, DHSIT and wireline services. This section of the paper describes each part of the section and their combination.

Over the past decade, acoustic telemetry has been successfully implemented in Drill Stem Testing (DST) operations, enabling real-time data transmission and remote actuation of downhole tools without the need for physical wiring (Temer et al., 2021). The technology functions by converting digital signals into acoustic waves, which are transmitted along the toolstring via a series of acoustic repeaters. These repeaters are strategically positioned to relay signals across various completion components—including packers, valves, plugs, and screens—ensuring continuous communication even in complex well configurations. The underlying physics of acoustic telemetry is based on the propagation of sound waves through solid media,

such as steel tubulars. Unlike electromagnetic signals, which are often attenuated in metallic and fluid-filled environments, acoustic waves maintain integrity over long distances in solid structures. The system encodes control commands and sensor data into high-frequency acoustic pulses, which are then decoded either at the surface or by downhole receivers. This enables real-time monitoring and control of downhole operations. To acquire gauge data in real time, one acoustic repeater must be connected to the downhole gauge, while another must interface with the surface acquisition system (Fig. 2). The success of this communication depends on the signal-to-noise ratio, which determines the clarity and reliability of the transmitted data.



Figure 2—Acoustic telemetry between two repeaters

Another key element is the DHSIT enabling the high-quality transient test data with minimum wellbore storage effect, which is critical to accurately estimate the reservoir and well parameters (Kuchuk, 1997). The DHSIT (Fig. 3) is conveyed in the well using slickline (or any other conveyance) and set downhole using a standard lock mandrel or retrievable packer. This provides an anchor and seal against the production string. The battery-powered, surface-programmable downhole actuator operates the multicycle instant shut-in valve. This generates a preprogrammed sequence of flowing and shut-in events. A memory pressure gauge, typically attached below the tool, records the borehole pressure during each step of the preprogrammed sequence. The acquired pressure data is then used for analysis and interpretation. At the end of the test sequence the valve is opened to equalize the pressure across the tool and prepare it for retrieval with slickline.

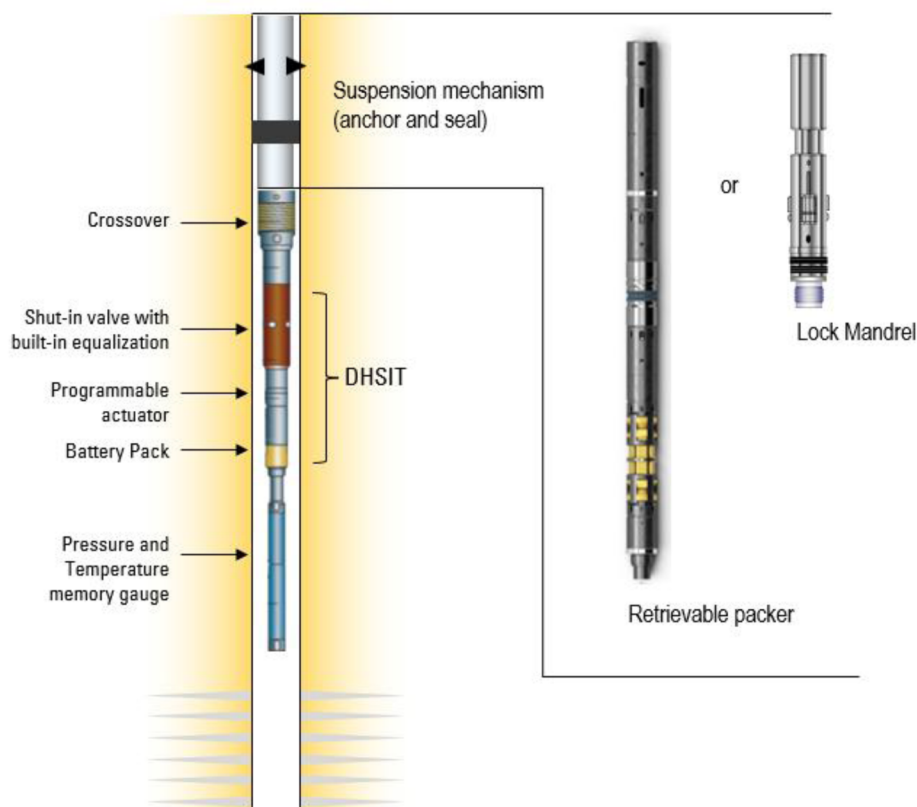


Figure 3—Downhole Shut-In Tool with a memory gauge

A critical component in achieving high-fidelity transient pressure data during rigless well testing is the Downhole Shut-In Tool (DHSIT). This tool plays a vital role in minimizing wellbore storage effects, which is essential for accurately estimating key reservoir and well parameters (Kuchuk, 1997). By isolating the pressure response closer to the reservoir, the DHSIT enables more precise interpretation of formation behavior, especially in early-time data. The DHSIT (Fig. 3) is typically deployed into the well using slickline, although other conveyance methods such as electric line or coiled tubing can also be used depending on operational requirements. Once at the target depth, the tool is anchored and sealed against the production tubing using a standard lock mandrel or a retrievable packer. This anchoring mechanism ensures tool stability and pressure integrity during the test. At the heart of the DHSIT is a battery-powered, surface-programmable actuator that controls a multicycle, instant shut-in valve. This valve is capable of executing a preprogrammed at surface before run in hole sequence of flow and shut-in events without surface intervention. A memory pressure gauge, usually mounted below the tool, continuously records downhole pressure throughout the test sequence. Upon completion of the test sequence based initially set timer, the valve is automatically opened to equalize pressure across the tool, allowing for safe retrieval via slickline. This autonomous operation not only reduces the need for real-time surface control but also enhances operational efficiency and safety in remote or rigless environments.

The third essential component in the rigless well testing setup is the wireline unit, which is typically available on-site and kept on standby during the pressure build-up phase. To enable seamless communication between downhole tools and surface systems, the technology team has developed a specialized adapter that interfaces the acoustic repeater with both the wireline cable and the surface data acquisition computer. This adapter is designed to accommodate various wireline cable configurations, including monocable, 2-32 cables, and even hepta-cables. The wireline included a weight bar to provide the necessary descent force and a gauge cutter to protect the repeater.

The real-time pressure data from the gauge positioned below the DHSIT is achieved by incorporating an acoustic repeater directly into the DHSIT assembly and deploying a second repeater via wireline during the pressure build-up phase (Fig. 4). This configuration establishes a communication link between the downhole gauge and the surface acquisition system without the need for physical contact or wiring. The acoustic repeater embedded in the DHSIT captures the historical and real-time pressure data from gauge and transmits it acoustically through the toolstring. The wireline-deployed repeater acts as a relay, receiving the acoustic signals at certain distance and forwarding them to the surface acquisition computer. This surface system is equipped to decode the signals and seamlessly integrate the data into the real-time data visualization platform.

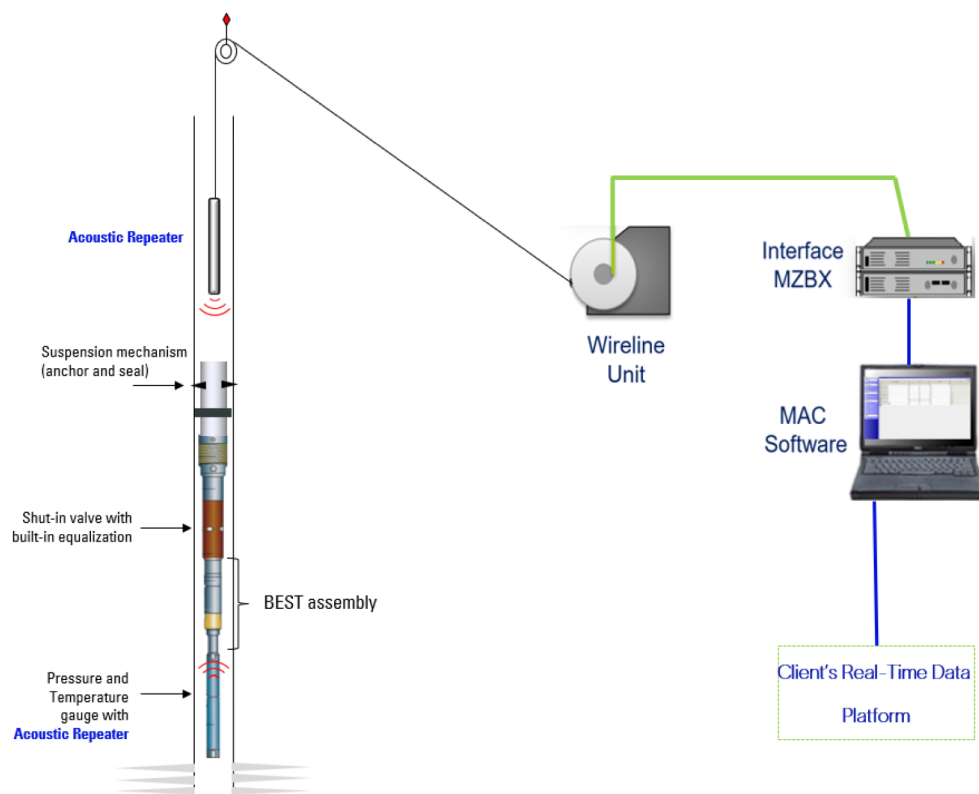


Figure 4—Real-time data from the gauge below DHSIT

Once the data reaches the visualization platform, petroleum engineers can access and interpret it in real time, enabling dynamic decision-making during the test. This capability is particularly valuable for optimizing build-up duration, identifying reservoir boundaries, and adjusting test parameters on the fly.

The reliability of this telemetry link depends on the signal-to-noise ratio, which must be carefully managed to ensure consistent and accurate data transmission. To improve the signal quality authors proposed adding a simple slickline centralizers on both assemblies, which increases the metal-to-metal contact which amplifies the signal-to-noise ratio to provide the seamless data transmission. During the deployment authors also evaluated the recommended distance between two repeaters.

Field validation and deployment

To validate the reliability and performance of the wireless telemetry system, a comprehensive integrity test was conducted at the base facility. The test setup was designed to simulate actual wellbore conditions and included all critical components: slickline deployment equipment, acoustic telemetry tools, and a wireline

unit. The main elements of the test assembly comprised a cable adapter, wireline acoustic repeater, weight bar, DHSIT equipped with a pressure gauge, and an integrated downhole repeater (Fig. 5).

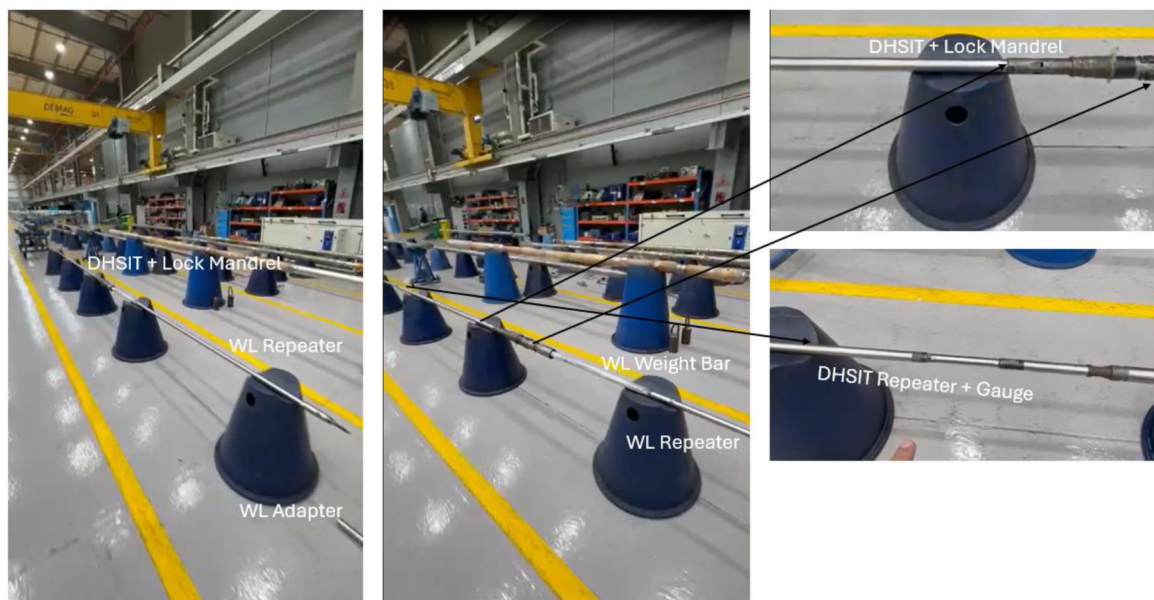


Figure 5—The system integrity test of the wireless telemetry in the base

During the test, the system successfully transmitted pressure data from the gauge below the DHSIT to the surface acquisition computer in real time. The data was displayed on the monitoring screen, confirming that communication between the downhole and surface components was established without any noticeable delay. Since the gauge reading atmospheric pressure, the team played with the temperature to observe the variation of the parameters in real-time. Additionally, the team explored the acoustic communication range by varying the distance between the two repeaters in open-air conditions. It was observed that reliable signal transmission could be achieved within a few meters, demonstrating the robustness of the acoustic telemetry system even outside the wellbore environment. These results affirm the system's potential for flexible deployment and its effectiveness in supporting rigless well testing operations.

The field deployment was achieved on 4 wells successfully proving the reliability of the wireless telemetry for rigless well testing. The field deployment confirmed the importance of the centralizer to improve the communication between two repeaters at higher distance.

Fig. 6 illustrates an example of bottomhole pressure (BHP) data, shown in purple, alongside a diagnostic log-log plot obtained in real time during one of the rigless well testing operations. In this case, the well was opened for a main flow period lasting 52 hours following the installation of the DHSIT equipped with a wireless pressure gauge. Approximately 10 hours after initiating the shut-in phase for the main build-up, the second acoustic repeater of the wireless telemetry system was deployed into the well using wireline. This repeater was positioned at a depth approximately 100 feet above the DHSIT. Once in place, communication was successfully established between the surface acquisition system and the gauge located below the DHSIT. Following the establishment of the telemetry link, the system began retrieving historical pressure data recorded during the main flow period. Depending on the volume of data, this retrieval process can take several hours. Once all historical data is uploaded, the system transitions automatically to real-time data streaming, allowing continuous monitoring of bottomhole pressure during 36 hours. The acquired pressure data is then transmitted to an interpretation software platform, where it is processed to generate diagnostic plots such as log-log and derivative curves. In this case the log-log plot exhibits the heterogeneity

in the reservoir at certain distance recommending to increase the build-up duration. If we observe an infinite-acting reservoir we could recommend to stop the build-up earlier.

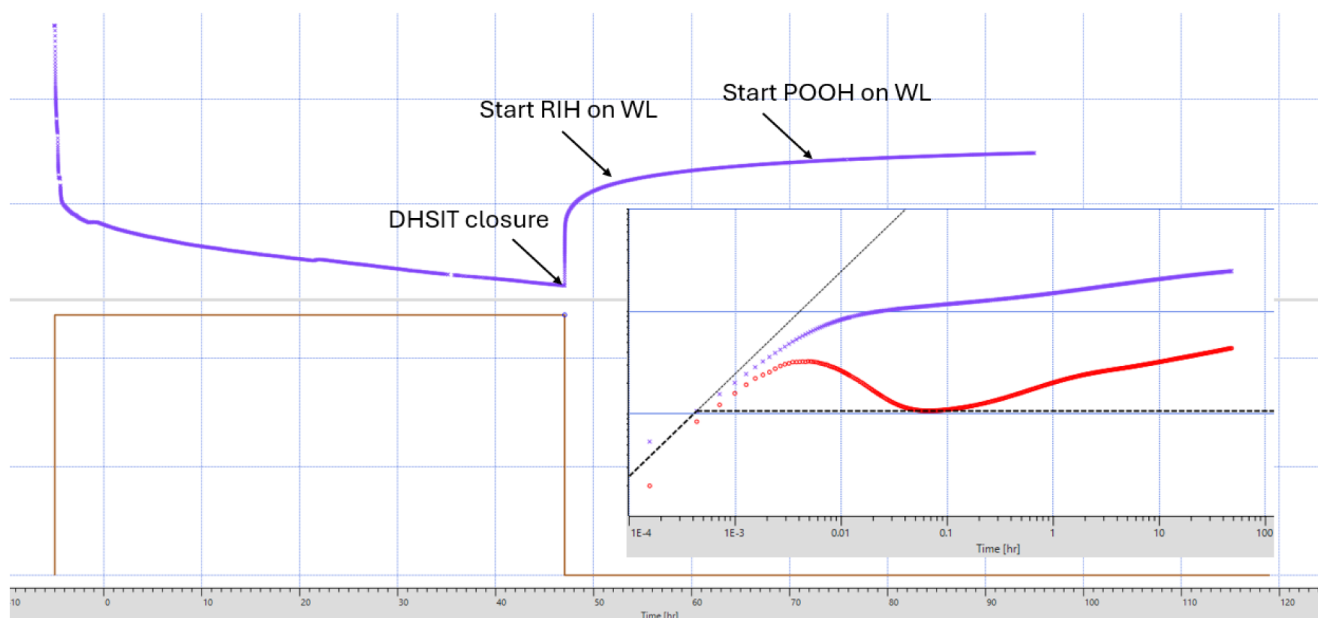


Figure 6—Field example #1 of the Wireless Telemetry for Rigless Well Testing

Fig. 7 presents the second field example, where we extracted 5 hours of flowing historical data prior to the pressure build-up and continued capturing bottomhole pressure data in real time. The diagnostic log-log plot (Fig. 7, right), generated in real time, revealed infinite-acting radial flow during the first 20 hours. This insight enabled us to shorten the build-up duration from 48 to 24 hours.

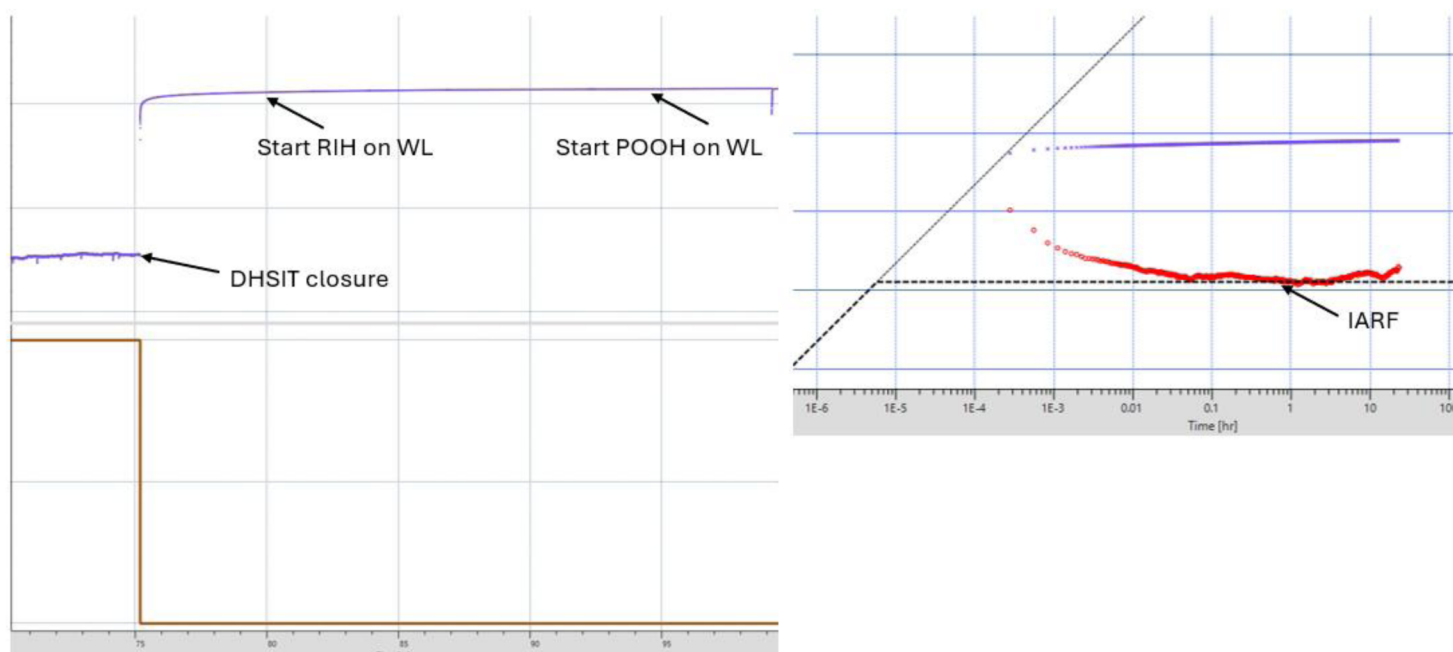


Figure 7—Field example #2 of the Wireless Telemetry for Rigless Well Testing

As part of the system validation process, the authors conducted a signal-to-noise ratio assessment during the retrieval of the wireline-deployed acoustic repeater. This test aimed to evaluate the robustness and

reliability of the wireless telemetry link at varying depths above the DHSIT. The procedure involved gradually pulling the repeater out of the hole and pausing at regular intervals—specifically every 200 feet—to monitor the quality of the acoustic signal. At each stop, the team assessed whether communication with the downhole gauge remained stable and whether real-time data continued to stream to the surface acquisition system. The results demonstrated consistent and uninterrupted signal transmission up to 1,200 feet above the DHSIT. This finding highlights the strength and resilience of the acoustic telemetry system, confirming its ability to maintain reliable communication over significant vertical distances within the wellbore. Such robustness is particularly valuable in rigless operations, where flexibility in tool deployment and retrieval is essential.

Way Forward and Recommendations

This project successfully demonstrated the feasibility of wirelessly communicating with a pressure gauge located below the (DHSIT), marking a significant milestone in the evolution of rigless well testing. Building on this achievement, the next phase of development focuses on expanding the system's functionality to include wireless control of the DHSIT itself. This enhancement would allow operators to interrupt or resume the pressure build-up phase remotely, eliminating the need for an additional slickline run with a mechanical prong to actuate the tool. The envisioned system will integrate wireless acoustic telemetry with digital slickline and coiled tubing technologies, creating a more versatile and efficient testing platform. This integration aims to streamline operations, reduce intervention time, and enhance safety by minimizing the number of runs into the well.

Beyond its immediate application in exploration well testing, the solution holds strong potential for broader deployment in production environments. By enabling real-time control and data acquisition during build-up periods, the system can help optimize test durations, reduce operational costs and provide earlier production in development wells. This scalability makes the technology a valuable asset for enhancing well testing strategies across the entire lifecycle of a field.

Conclusion

In summary, the authors introduced the new solution to optimize the rigless well testing and enhance the pressure transient test for better reservoir characterization. It was done in a collaboration with the technology and operational team and included the proof of concept, the base and field validation and improvement.

Nomenclature

BHP	– Bottomhole pressure
DHSIT	– Downhole Shut-In Tool
DST	– Drill Stem Test
SNR	– Signal to Noise Ratio

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